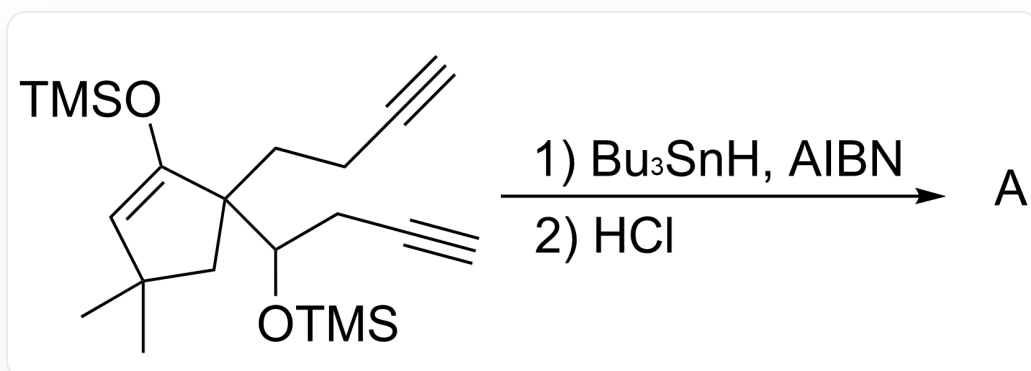
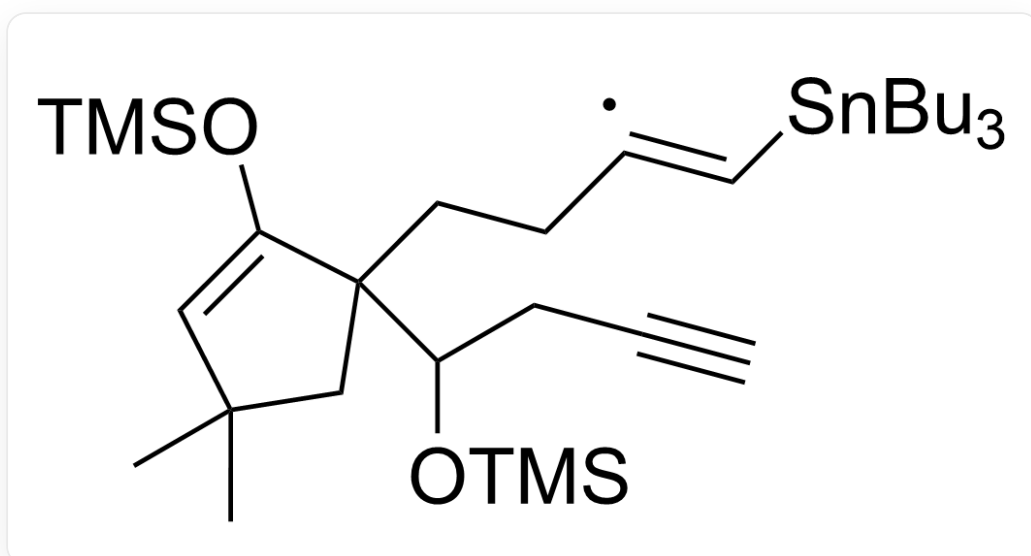


Question

Radical rearrangement reactions have wide applications in organic synthesis.

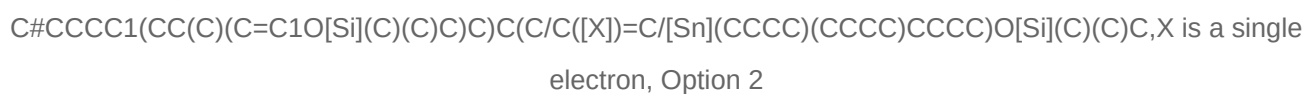


C#CCCC1(C(O[Si](C)(C)C)CC#C)CC(C)(C)C=C1O[Si](C)(C)C>[Bu_3SnH].[AIBN]>[A], A is the reaction product. After the reaction is complete, HCl is added.

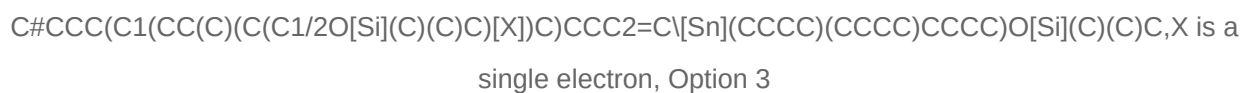


C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C)C(C/C([X])=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C, X is a single electron, Option 1

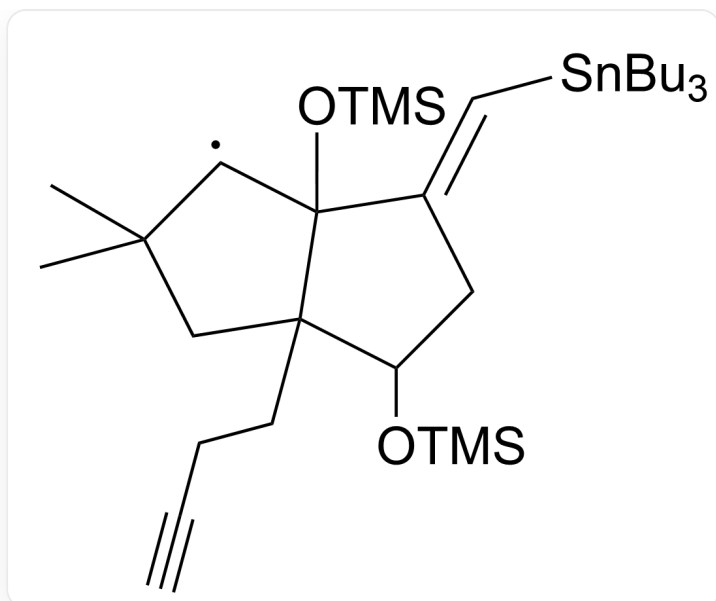
Option 1



Option 2

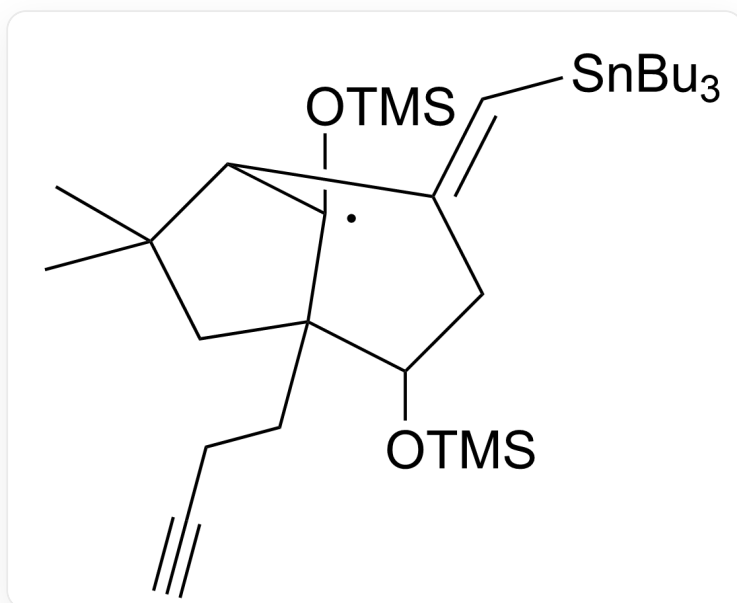


Option 3



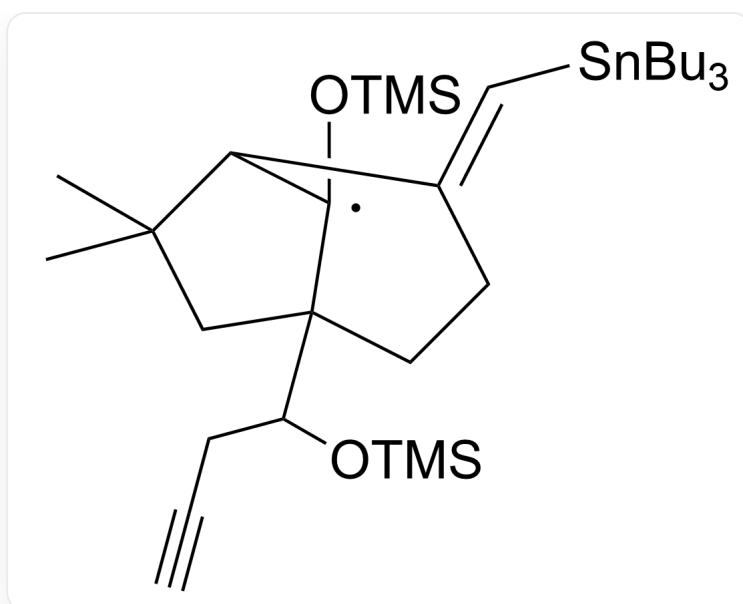
CC1(C([X])C2(C(CCC#C)(C1)C(C/C2=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C)C, X is a single electron, Option 4

Option 4



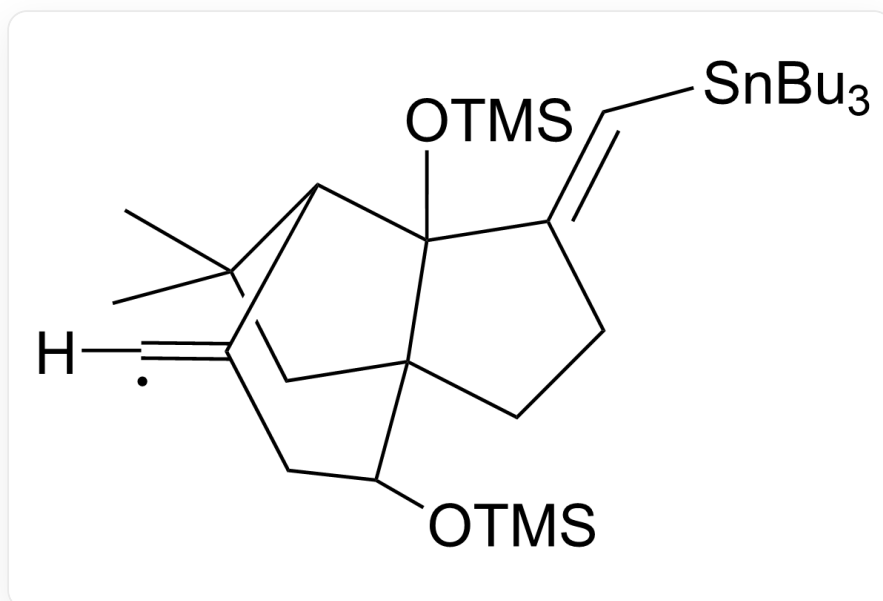
CC1(C2C(C(CCC#C)(C1)C(C/C2=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)([X])O[Si](C)(C)C)C, X is a single electron, Option 5

Option 5



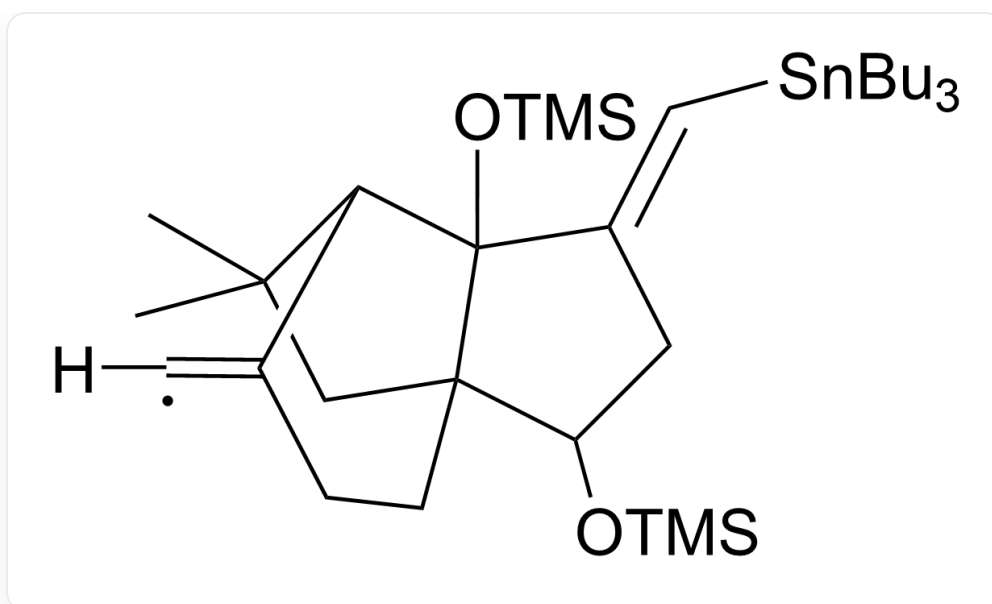
CC1(C2C(C(C(CC#C)O[Si](C)(C)C)(C1)CC/C2=C\[Sn](CCCC)(CCCC)CCCC)([X])O[Si](C)(C)C)C,X is a single electron, Option 6

Option 6



CC1(C2C3(C(C(C/C2=C([X])\[H])O[Si](C)(C)C)(C1)CC/C3=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)C,X is a single electron, Option 7

Option 7



CC1(C2C3(C(CC/C2=C([X])\[H]))(C1)C(C/C3=C\[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C)C,X
 is a single electron, Option 8

Option 8

Without considering enantiomers, please select all possible intermediates in the options.

A. All other options are incorrect

B. Option 1, 3, 5

C. Option 2, 4, 5

D. Option 1, 3, 7

E. Option 2, 4, 8

F. Options 1, 3, 6

G. Option 2, 5

H. Options 2, 5, 6

I. Options 1, 4, 6

Answer

Correct Answer: **F**

Detailed Explanation

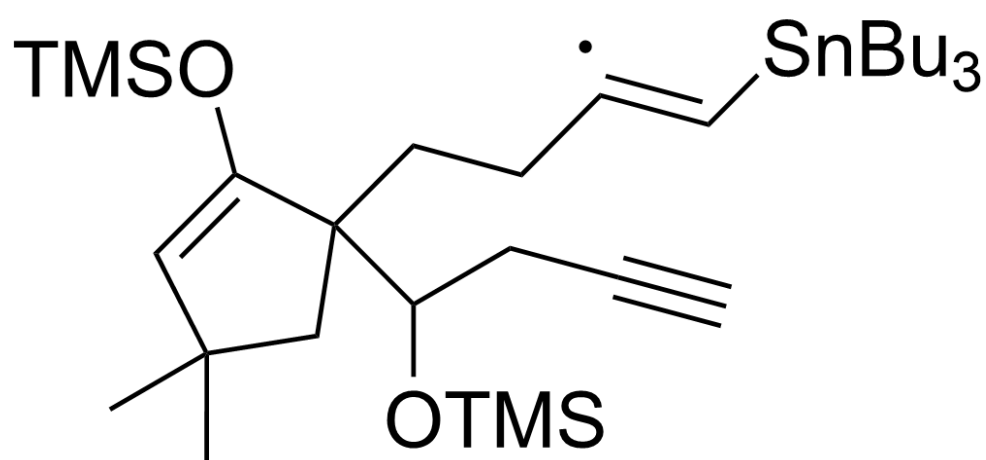
The carbon radical with the electron-withdrawing group $-OTMS$ introduced at the γ position is relatively unstable.

CHECKPOINT

1 PTS

The carbon radical with the electron-withdrawing group $-OTMS$ introduced at the γ position is relatively unstable

First, an intermediate is formed.



C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C(C/C([X])=C/[Sn](CCCC)(CCCC)CCCCO[Si](C)(C)C,X为单电子

CHECKPOINT

1 PTS

First, an intermediate is formed: C#CCCC1(CC(C)(C=C1O[Si](C)(C)C)C(C/C([X])=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C, X is single electron

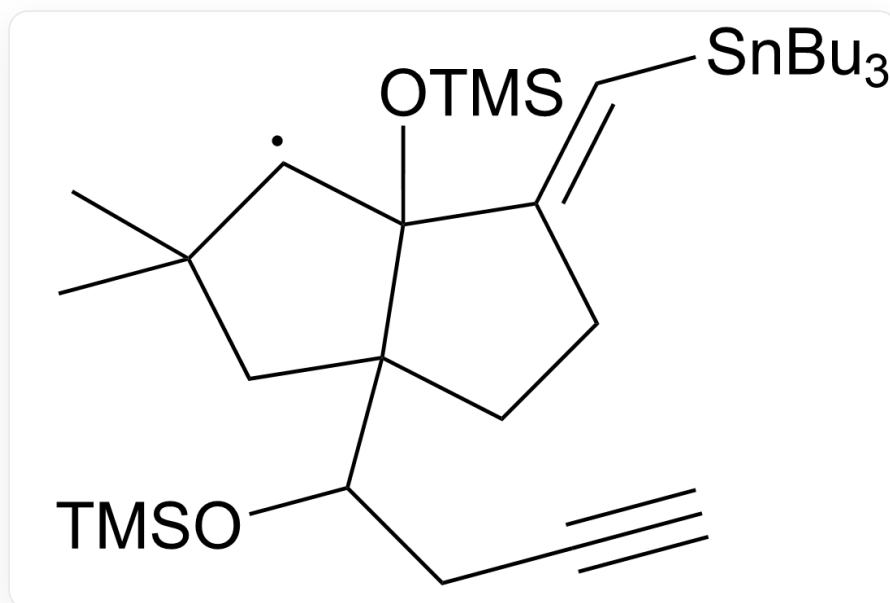
The reaction rate of 5-exo-trig is faster.

CHECKPOINT

1 PTS

The reaction rate of 5-exo-trig is faster

Therefore, the intermediate is preferentially generated.



CC1(C([X])C2(C(CCC#C)(C1)C(C/C2=C/[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C, X is single electron

CHECKPOINT

1 PTS

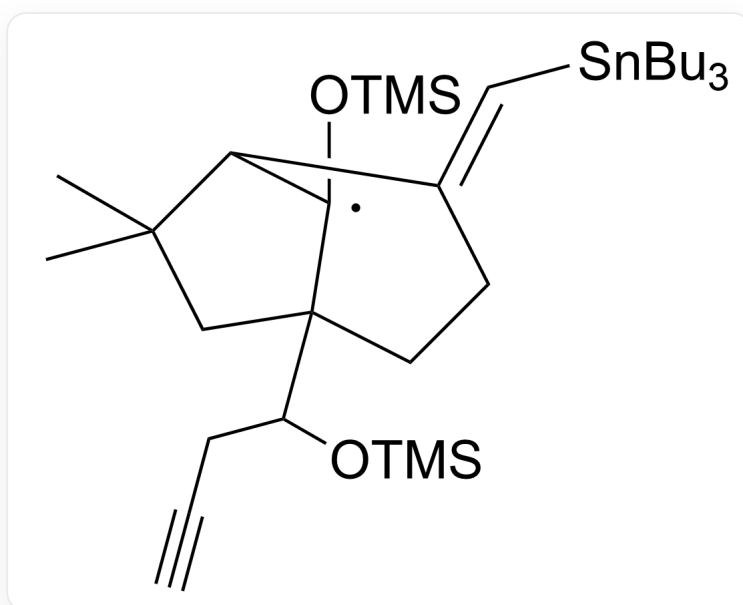
Next, the intermediate is preferentially formed: CC1(C([X])C2(C(CCC#C)(C1)C(C/C2=C[Sn](CCCC)(CCCC)CCCC)O[Si](C)(C)C)O[Si](C)(C)C)C, X is single electron

Then, a fast 1,2-rearrangement reaction yields a carbon radical stabilized by n electrons of oxygen.

CHECKPOINT

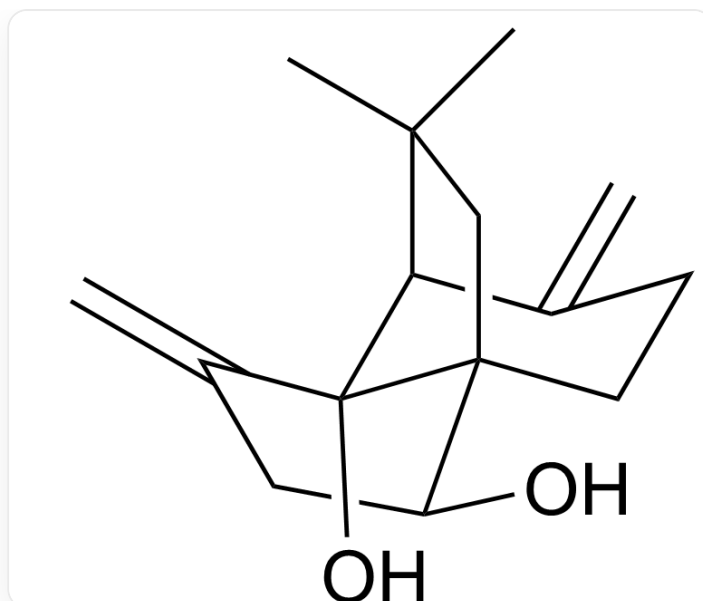
1 PTS

Then, a fast 1,2-rearrangement reaction yields a carbon radical stabilized by n electrons of oxygen



CC1(C2C(C(C(CCC#C)O[Si](C)(C)C)(C1)CC/C2=C[Sn](CCCC)(CCCC)CCCC)([X])O[Si](C)(C)C)C, X是单电子

Finally, the obtained product A is



CC1(C)[C@@H]2[C@@](C(C[C@@H]3O)=C)(O)[C@]3(C1)CCC2=C